

A TOPOLOGICAL FIELD THEORY FOR NON-ABELIAN VORTICES

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We construct a two-dimensional topological field theory for non-abelian Higgs vortices. We discuss relevant features of the resulting BRST quantized theory and also discuss topological invariants.

1. Introduction

The interest in topological field theory (TFT) has been constantly growing since Witten's works on supersymmetric Yang–Mills [1], sigma [2] and Chern–Simons [3] models. The relevance of TFTs stems both from their mathematical and physical implications [4–17]. On the one hand TFTs give a quantum field theoretical method of arriving at instanton moduli space and constructing topological invariants – Donaldson invariants, Jones polynomials, etc. On the other hand topological models may prove to be valuable tools in understanding fundamental physics, in particular Chern–Simons gauge theories in $d=3$ dimensions are intimately related to rational conformal field theories in $d=2$ dimensions [3,14–17].

As first observed by Baulieu and Singer [4], TFTs can be regarded as topological actions that must be gauge fixed with respect to the diffeomorphism group. Another point of view is that the action of a TFT is the BRST quantization of the Langevin equation that one may associate with the instanton defining equation [5–7]. Following this last approach, an example of TFT with a Higgs scalar field providing an explicit gauge symmetry breaking was presented by one of us and Thompson [11] (previous examples in this context correspond to the Prasad–Sommerfield limit of a $d=3$ Georgi–Glashow model in which no $\lambda\phi^4$ coupling is present [6,12]).

The abelian model presented in ref. [11] showed interesting connections between TFTs, SUSY Higgs model and vortex properties – superconducting vortices are the two-dimensional instantons of this model and will dominate its semiclassical limit. A topological quantum field theory of vortices has also been studied by Chapline and Grossman [13] starting from a topological invariant action and quantizing it à la Baulieu–Singer [4]. In this way, a link between conformal field theory and Donaldson theory of four-manifolds was set up.

We discuss in this letter a topological field theory for non-abelian vortices (instantons). As was established in ref. [18], the non-abelian Higgs model has first order instanton equations, the Bogomol'nyi equations, when a particular relation between coupling constants holds. Moreover, exact solutions to the resulting system are known to exist. Following the approach in refs. [5,6], we start from these equations and construct a two-dimensional TFT which corresponds to the non-abelian Higgs model. In fact, using the Batalin–Vilkovisky algorithm [19] we proceed to the BRST quantization of the Langevin equation associated to the Bogomol'nyi equations arriving in this way to a topological field theory. We end this work by discussing properties of the model and the topological invariants that can be constructed.

2. Vortices in non-abelian gauge theories

As is well known vortex configurations exist when-

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ever the gauge symmetry is spontaneously broken via Higgs fields [20], leaving invariant the vacuum under a certain subgroup H of the gauge group G . The magnetic flux is quantized and this is intimately related to the non-triviality of the relevant homotopy group.

For $G = SU(N)$, vortex solutions have been found for maximum symmetry breaking [20]. This can be achieved taking N Higgs fields in the adjoint representation and choosing an adequate symmetry breaking potential. In this case $H = Z_N$ and $\Pi_1(G/H) = Z_N$. One then has $N-1$ topologically inequivalent configurations (compare with the abelian case, $G = U(1)$, $H = I$, $\Pi_1(G/H) = Z$).

Following this scheme we have found in ref. [18] the first-order system of Bogomol'nyi equations for non-abelian vortices. For the $SU(2)$ case they can be written as

$$\begin{aligned} E &\equiv \frac{1}{2} \epsilon_{\mu\nu} F_{\mu\nu} - \psi \operatorname{tr}(\phi^2 - \phi_0^2) = 0, \\ E_\mu &\equiv D_\mu^+ \phi - i[\psi, D_\mu^+ \phi] = 0, \\ F_\mu &\equiv D_\mu^+ \psi = 0, \end{aligned} \quad (1)$$

where the gauge field A_μ takes values in the $SU(2)$ algebra, ϕ and ψ are two Higgs fields in the adjoint representation, D_μ is the covariant derivative and

$$D_\mu^+ \equiv \frac{1}{2} (D_\mu + i\epsilon_{\mu\nu} D_\nu). \quad (2)$$

We work in a two-dimensional euclidean spacetime and we take $\epsilon_{01} = 1$.

System (1) arises from an action in which one of the Higgs fields (ψ) plays no dynamical role. Its presence ensures complete symmetry breaking, a necessary condition in order to have non-trivial vortices [20].

As is well known [18], Bogomol'nyi equations are valid for particular relations between coupling constants. For non-abelian vortices, system (1) holds provided the Higgs coupling constant, λ_ϕ and the gauge coupling constant e satisfy the relation $\lambda_\phi = e^2$. Concerning the topological charge of the Z_2 vortices satisfying system (1), it is given, modulo Z_2 , by the following expression:

$$Q = \frac{1}{2\pi} \operatorname{tr} \int d^2x \psi \epsilon_{\mu\nu} F_{\mu\nu}. \quad (3)$$

3. The topological model

In order to construct a topological field model for non-abelian vortices we consider the following classical action:

$$S_{\text{cl}} = \int d^2x (|G - E|^2 + |G_\mu - E_\mu|^2 + |H_\mu - F_\mu|^2), \quad (4)$$

where G , G_μ and H_μ are the random fields belonging to the $SU(2)$ algebra. G_μ and H_μ are self-dual (with our conventions a self-dual field verifies $K_\mu = i\epsilon_{\mu\nu} K_\nu$).

The action (3) is invariant under the following infinitesimal transformations:

$$\begin{aligned} \delta A_\mu &= \epsilon_\mu - D_\mu \omega, \\ \delta \phi &= \lambda - [\phi, \omega], \quad \delta \psi = \eta - [\psi, \omega], \end{aligned} \quad (5a)$$

$$\delta G = \epsilon_{\rho\mu} D_\rho \epsilon_\mu - 2\psi \operatorname{tr}(\phi \lambda) - \psi \operatorname{tr}(\phi^2 - \phi_0^2) - [G, \omega], \quad (5b)$$

$$\begin{aligned} \delta G_\mu &= (\delta_{\mu\nu} + i\epsilon_{\mu\nu}) (-[\phi, \epsilon_\nu] - \epsilon_{\nu\rho} [\psi, [\phi, \epsilon_\rho]] + D_\nu \lambda \\ &\quad + \epsilon_{\nu\rho} [\psi, D_\rho \lambda] - \epsilon_{\nu\rho} [D_\rho \phi, \eta]) - [G_\mu, \omega], \end{aligned} \quad (5c)$$

$$\delta H_\mu = (\delta_{\mu\nu} + i\epsilon_{\mu\nu}) (-[\psi, \epsilon_\nu] + D_\nu \eta) - [H_\mu, \omega]. \quad (5d)$$

The basic transformation laws are those given by eqs. (5a). They are supplemented with transformations (5b)–(5d) for the auxiliary fields chosen so as to ensure invariance of S_{cl} . In eqs. (5a) ω is the usual gauge parameter while ϵ_μ , λ and η correspond to arbitrary variations of the physical fields A_μ , ϕ and ψ . The splitting in two terms may look somehow artificial since the terms corresponding to usual gauge transformations can be included through a particular choice of the parameters ϵ_μ , λ and η . It is interesting to note that the theory can also be quantized if terms containing ω are dropped from transformations (5a) but maintained in transformations (5b)–(5d) where they represent a truly independent symmetry. In any case the resulting quantum action is the same (see below).

Transformations (5) are just enough to choose a gauge in which G , G_μ and H_μ vanish. This can be achieved by using parameters ϵ_μ , λ and η , leaving untouched parameter ω . Although auxiliary fields have three independent components (G_μ and H_μ are self-dual fields) each one taking values in the $SU(2)$ algebra while parameters ϵ_μ , λ and η are four, not all of

them are effective regarding gauge transformations. Indeed, the classical action (4) possesses an on-shell invariance: if one chooses ϵ_μ , λ , η and ω satisfying

$$\epsilon_\mu = D_\mu A, \quad \omega = A, \quad \lambda = -[A, \phi], \quad \eta = -[A, \psi], \quad (6)$$

then all fields remain invariant under transformations (5).

Finally it is important to note that in the gauge $G=0$, $G_\mu=H_\mu=0$ the equations of motion associated to S_{cl} coincide with the Bogomol'nyi equations.

We are now ready to quantize the theory defined by S_{cl} . In view of the above discussed features of its symmetries, the naive Faddeev-Popov procedure cannot be applied. We will instead use the Batalin-Vilkovisky method [19] specialized to the case of a first stage theory (which corresponds in their classification to the second generation invariance observed in eq. (6)).

We then start by adding to the original fields $\mathbb{A}^i \equiv (A_\mu, \phi, \psi, G, G_\mu, H_\mu)$ ghost fields $C^a \equiv (\psi_\mu, c, \rho, \xi)$ and σ associated to each one of the symmetries of S_{cl} . All ghost fields entering in C^a are Grassmann odd and σ is Grassmann even.

$$\epsilon_\mu \rightarrow \psi_\mu, \quad \omega \rightarrow c, \quad \lambda \rightarrow \rho, \quad \eta \rightarrow \xi, \quad A \rightarrow \sigma. \quad (7)$$

The ghost number of the minimal set $\Phi_{min} = (\mathbb{A}^i, C^a, \sigma)$ is (0, 1, 2). For each field in Φ_{min} an antifield of opposite statistics and with ghost number minus the ghost number of the field minus one is then introduced. These antifields are $\Phi_{min}^* = (\mathbb{A}^{i*}, C^{a*}, \sigma^*)$ and have ghost number $(-1, -2, -3)$. One then constructs an action-like object $S_{min}[\Phi_{min}, \Phi_{min}^*]$, such that $S_{min}[\Phi_{min}, \Phi_{min}^* = 0] = S_{cl}$, which has ghost number 0 and satisfies the master equation

$$\frac{\delta_r S}{\delta \Phi^i} \frac{\delta_l S}{\delta \Phi^{i*}} = 0, \quad (8)$$

subscript r (l) indicates right (left) derivatives. Moreover, S in eq. (8) must satisfy the following boundary conditions:

$$\left. \frac{\delta_l \delta_r S}{\delta \mathbb{A}^i \delta C^a} \right|_{\Phi^* = 0} = R^i_a, \quad \left. \frac{\delta_l \delta_r S}{\delta C^a \delta \sigma} \right|_{\Phi^* = 0} = Z^a, \quad (9)$$

R^i_a are defined from transformations $\delta \mathbb{A}^i = R^i_a \theta^a$, $\theta = (\epsilon_\mu, \lambda, \eta, \omega)$ and Z_a satisfies the zero-mode equation $R^i_a Z^a = 0$, on-shell.

As gauge conditions we choose

$$\partial_\mu A_\mu = 0, \quad D_\mu \psi_\mu = 0, \quad G_\mu = H_\mu = 0, \quad G = 0. \quad (10)$$

The "gauge fermion" Ψ enforcing these conditions is

$$\Psi = \bar{c} \partial_\mu A_\mu + \bar{\sigma} D_\mu \psi_\mu + XG + X_\mu G_\mu + \tau_\mu H_\mu, \quad (11)$$

where new ghost fields $\tilde{C}^a \equiv (\bar{c}, X, X_\mu, \tau_\mu)$ and $\bar{\sigma}$ with ghost number -1 , Grassmann odd and -2 , Grassmann even, respectively, have been introduced (fields X_μ and τ_μ are self-dual). One also introduces Lagrange multipliers a, d, d_μ, e_μ with ghost number 0, Grassmann even and b with ghost number -1 , Grassmann odd. All these fields and their corresponding antifields are used to construct a new term ΔS which is added to the minimal action in such a way that the complete action still satisfies the master equation

$$S[\Phi, \Phi^*] = S_{min}[\Phi, \Phi^*] + \Delta S[\Phi, \Phi^*], \quad (12)$$

$$\begin{aligned} \Delta S = & \int d^2x \{ \bar{c}^* a + \bar{\sigma}^* b + X^* (d - [X, c]) \\ & + X_\mu^* (d_\mu - [X_\mu, c]) + \tau_\mu^* (e_\mu - [\tau_\mu, c]) \\ & - d^* ([d, c] + [X, \sigma]) - d^* ([d, c] + [X, \sigma]) \\ & - e^* ([e, c] + [\tau, \sigma]) \}. \end{aligned} \quad (13)$$

The final quantum action S_Q is obtained on restricting the antifields to lie on the surface $\Phi^* = \delta \Psi / \delta \Phi$. If one integrates over G, G_μ and H_μ one arrives at the final form of S_Q :

$$\begin{aligned} S_Q = & \int d^2x \{ [\tfrac{1}{2} \epsilon_{\mu\nu} F_{\mu\nu} - \psi \text{tr}(\phi^2 - \phi_0^2)] d \\ & - \tfrac{1}{4} d^2 + (D_\mu^+ \phi - i[\psi, D_\mu^+ \phi]) d_\mu - \tfrac{1}{4} d_\mu^2 + D_\mu^+ \psi e_\mu \\ & - \tfrac{1}{4} e_\mu^2 + X(\epsilon_{\rho\mu} D_\rho \psi_\mu - 2\psi \text{tr}(\phi, \rho)) \\ & - (\text{tr}(\phi^2 - \phi_0^2) \xi) + \tfrac{1}{4} [X, X] \sigma + X_\mu (\delta_{\mu\nu} + i\epsilon_{\mu\nu}) \\ & \times (-[\phi, \psi_\nu] + D_\nu \rho - \epsilon_{\nu\rho} [\psi, [\phi, \psi_\rho]]) \\ & + \epsilon_{\nu\rho} [\phi, D_\rho \rho] - \epsilon_{\nu\rho} [D_\rho \phi, \xi] \\ & + \tfrac{1}{4} [X_\mu, X_\mu] \sigma + \tau_\mu (\delta_{\mu\nu} + i\epsilon_{\mu\nu}) (-[\psi, \psi_\nu] + D_\nu \xi) \\ & + \tfrac{1}{4} [\tau_\mu, \tau_\mu] \sigma + (-\partial_\mu \bar{c} + [\psi_\mu, \sigma]) (\psi_\mu - D_\mu c) \\ & + (\partial_\mu A_\mu) a + (D_\mu \psi_\mu) b \}. \end{aligned} \quad (14)$$

S_Q is invariant under the BRST transformation which can be computed using

$$\delta\Phi^i \equiv \alpha\{Q, \Phi^i\} = -\alpha \frac{\delta S}{\delta\Phi_i^*}, \quad (15)$$

where α is an infinitesimal anticommuting parameter. The field transformations are

$$\begin{aligned} \delta A_\mu &= \alpha(\psi_\mu - D_\mu c), \quad \delta\phi = \alpha(\rho - [\phi, c]), \\ \delta\psi &= \alpha(\xi - [\psi, c]), \quad \delta\psi_\mu = -\alpha(D_\mu\sigma + [c, \psi_\mu]), \\ \delta c &= -\alpha(\sigma + \tfrac{1}{2}[c, c]), \quad \delta\rho = -\alpha([\phi, \sigma] + [c, \rho]), \\ \delta\xi &= -\alpha([\psi, \sigma] + [c, \xi]), \quad \delta\sigma = \alpha[\sigma, c], \\ \delta\bar{c} &= \alpha a, \quad \delta\bar{c} = \alpha b, \\ \delta X &= \alpha(d - [X, c]), \quad \delta d = \alpha([d, c] + [X, \sigma]), \\ \delta X_\mu &= \alpha(d_\mu - [X_\mu, c]), \\ \delta d_\mu &= \alpha([d_\mu, c] + [X_\mu, \sigma]), \\ \delta\tau_\mu &= \alpha(e_\mu - [\tau_\mu, c]), \quad \delta e_\mu = \alpha([e_\mu, c] + [\tau_\mu, \sigma]). \end{aligned} \quad (16)$$

It is important to stress that transformations (16) are nilpotent off-shell.

Although (15) looks rather complicated, it is easy to see that it can be written as the BRST transformed of a functional V ,

$$S_Q = \{Q, V\}, \quad (17)$$

with V given by

$$\begin{aligned} V = \int d^2x \{ & X[\tfrac{1}{4}d - \tfrac{1}{2}e_{\mu\nu}F_{\mu\nu} - \psi \operatorname{tr}(\phi^2 - \phi_0^2)] \\ & + X_\mu(\tfrac{1}{4}d_\mu - D_\mu^+\phi + i[\psi, D_\mu^+\phi]) + \tau_\mu(\tfrac{1}{4}e_\mu - D_\mu^+\psi) \\ & - \bar{c}\partial_\mu A_\mu - \bar{\sigma}D_\mu\psi_\mu \}. \end{aligned} \quad (18)$$

From (17) one trivially has $\{Q, S_Q\} = 0$.

We then see that the TFT defined by action S_Q belongs to the class of theories which complete action can be written as the BRST transformed of some functional (in contrast with those theories like the Chern–Simons one, in which this is not possible although the resulting generating functional is still metric independent).

The vacuum expectation value (VEV) of the energy–momentum tensor $T_{\mu\nu}$,

$$T_{\mu\nu} = \{Q, g^{-1/2}\delta V/\delta g^{\mu\nu}\} \quad (19)$$

vanishes,

$$\langle T_{\mu\nu} \rangle = 0, \quad (20)$$

since the VEV of any BRST transformations is zero. Note that integrating over Lagrange multipliers d, d_μ and e_μ one obtains the usual non-abelian Higgs action (at the critical point) plus the topological term (3), plus ghost terms.

Expressions (19), (20) mean that the partition function

$$Z = \int \mathcal{D}\text{fields} \exp[-(1/e^2)S_Q] \quad (21)$$

is a topological invariant. As stressed by Witten [1], for similar reasons Z is independent of coupling constants. Indeed, due to eqs. (14)–(17)

$$\frac{\delta Z}{\delta e^2} = 0 \quad (22)$$

for $e^2 \neq 0$. Therefore, one evaluates Z by going to the limit of small e^2 , where the path integral is dominated by classical minima which in the present case correspond to the solutions of Bogomol'nyi equations (1). Since these solutions are known to exist and have been explicitly constructed in ref. [20], one can hope to have a complete understanding of the theory by performing an instanton expansion. This is particularly attractive in view of the connections between two-dimensional TFTs associated to vortices and supersymmetric and superstring theories.

We end this work by constructing topological invariants associated to the action S_Q . In order to find operators W_k from which topological invariants can be constructed we consider the zig-zag equations

$$\begin{aligned} \{Q, W_0\} &= 0, \\ W_0 &= \tfrac{1}{2} \operatorname{tr} \sigma^2, \quad dW_0 = \{Q, W_1\}, \\ W_1 &= \operatorname{tr} \sigma \psi_\mu dx^\mu, \quad dW_1 = \{Q, W_2\}, \\ W_2 &= \operatorname{tr} F_{\mu\nu} \sigma dx^\mu \wedge dx^\nu dW_2, \quad dW_2 = 0. \end{aligned} \quad (23)$$

Following Witten's arguments one can easily prove that $\langle W_0 \rangle$ is independent on the point in space in which it is evaluated and corresponds to a topological invariant of the theory. W_2 is associated to the Z_2 topological charge of vortex configurations. In fact, note that W_2 has an expression analogous to (3) except that in the former case the ghost σ replaces the Higgs field ψ in defining the topological charge. From

W_k , integrals $I(\gamma) = \int_{\gamma} W_k$ over k -dimensional homology cycles can be constructed such that

$$I(\gamma) = i \left\{ Q, \int_{\beta} W_{k+1} \right\}$$

($\gamma = \partial\beta$). Then, VEVs of products of an adequate number of integrals $I(\gamma)$ correspond to topological invariants (see ref. [1] for details).

In summary we have shown that the TFT for vortices of a non-abelian gauge theory with Higgs fields can be constructed starting from Bogomol'nyi equations and quantizing the theory à la Batalin and Vilkovisky. The corresponding quantum action (14) has a rich structure which cannot be inferred by dimensional reduction of Witten's model in four dimensions. In the gauge we worked, it corresponds to the action of a non-abelian Higgs model plus a topological term, plus ghost terms. Because of the ability to get around the standard spin-statistics theorem in two dimensions a supersymmetric theory should arise once BRST ghosts are adequately written as spinors. This, as well as the connections of this theory with other two- and three-dimensional models (in particular CFTs in two dimensions) will be discussed elsewhere.

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